King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 233

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The volume of the solid generated by revolving the region bounded by the graphs of $y = x^2$ and $y = 2x - x^2$ about the x-axis is

(a) $\dfrac{\pi}{2}$

(b) 6π

(c) 3π

(d) 2π

(e) $\dfrac{\pi}{3}$

2. The volume of the solid generated by revolving the region bounded by the graphs of $x = y^2$ and $x = 4$ about the line $x = 5$ is given by the integral:

(a) $2\pi \int_0^2 (14 - 5y^2 + y^4)\, dy$

(b) $\pi \int_{-2}^2 (y^2 - 4)\, dy$

(c) $2\pi \int_0^2 (24 - 10y^2 + y^4)\, dy$

(d) $2\pi \int_0^2 (y^4 - 1)\, dy$

(e) $\pi \int_{-2}^2 (5 - y^2)^2\, dy$

3. $\int_{-1}^1 \operatorname{sech}^3 x \tanh x\, dx =$

(a) $\dfrac{2e^3 - 16}{3e^3}$

(b) $\dfrac{3e^3 + 16}{2e^3}$

(c) $\dfrac{3e^3 - 16}{2e^3}$

(d) 0

(e) $\dfrac{2e^3 + 16}{3e^3}$

4. $\int_0^{\pi} \left(1 - \tan\frac{\theta}{4}\right) d\theta =$

(a) $\pi - \ln 2$

(b) $\pi - 4\ln 2$

(c) $\pi + 2\ln 2$

(d) $\pi - 2\ln 2$

(e) $\pi + 3\ln 2$

5. Let f be a continuous function and c be a positive real number such that

$$\int_c^x f(t)\,dt = x^2 + 2x - 3.$$

Then $f(c) =$

(a) 2

(b) 4

(c) 5

(d) 6

(e) 3

6. $\int_3^4 \frac{3x}{(x-2)^2}\,dx =$

(a) $3 + \ln 2$

(b) $3 + 3\ln 2$

(c) $2 + 2\ln 2$

(d) $2 + 3\ln 2$

(e) $3 + 2\ln 2$

7. $\int_0^3 \sqrt{9 - x^2}\,dx =$

(a) $\frac{9}{2}\pi$

(b) 9π

(c) $\frac{9}{8}\pi$

(d) 18π

(e) $\frac{9}{4}\pi$

8. The average value of the function

$$f(x) = \frac{4(x^2+1)}{x^2}$$

over the interval $[1, 4]$ is

(a) 6

(b) 12

(c) 10

(d) 5

(e) 14

9. $\int_0^{1/\sqrt{2}} \frac{\cos^{-1}(x)}{\sqrt{1-x^2}}\, dx =$

(a) $\frac{3\pi^2}{32}$

(b) $\frac{3\pi^2}{16}$

(c) $\frac{\pi^2}{32}$

(d) $-\frac{3\pi^2}{16}$

(e) $-\frac{3\pi^2}{32}$

10. $\int_3^{\sqrt{3}} \frac{dx}{x\sqrt{4x^2-9}} =$

(a) $\frac{\pi}{18}$

(b) $-\frac{\pi}{9}$

(c) $-\frac{\pi}{18}$

(d) $\frac{\pi}{6}$

(e) $\frac{\pi}{9}$

11. If a and b are the real numbers that maximize $\int_a^b (4 - x^2)\,dx$, then $a + 2b =$ (Hint: Graph the function $f(x) = 4 - x^2$.)

(a) 2

(b) 0

(c) 4

(d) -4

(e) -2

12. $\int_9^{16} \dfrac{10}{\sqrt{x}\,(1+\sqrt{x})^2}\,dx =$

(a) 0

(b) 1

(c) 3

(d) 2

(e) 4

13. The area of the region bounded by the graphs of $x = 4 - y^2$ and $x = y + 2$ is equal to

(a) $\dfrac{9}{4}$

(b) $\dfrac{9}{2}$

(c) $\dfrac{15}{2}$

(d) $\dfrac{7}{2}$

(e) $\dfrac{15}{4}$

14. $\int 15x\sqrt{1-x}\,dx =$

(a) $10(1-x)^{5/2} - 6(1-x)^{3/2} + C$

(b) $4(1-x)^{5/2} - 6(1-x)^{3/2} + C$

(c) $4(1-x)^{5/2} - 10(1-x)^{3/2} + C$

(d) $6(1-x)^{5/2} - 10(1-x)^{3/2} + C$

(e) $6(1-x)^{5/2} - 4(1-x)^{3/2} + C$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Washers on $[0,1]$: $\pi\int_0^1(4x^2-4x^3)dx=\frac{\pi}{3}$.
2	**(c)**	Washers: $R=5-y^2$, $r=1$; $V=2\pi\int_0^2(24-10y^2+y^4)dy$.
3	**(d)**	Odd integrand on $[-1,1]$, so the integral is 0.
4	**(d)**	$\int\tan\frac{\theta}{4}d\theta=4\ln\sec\frac{\theta}{4}$, giving $\pi-4\ln\sqrt{2}=\pi-2\ln 2$.
5	**(b)**	$x=c$ gives $c=1$; $f(x)=2x+2$, so $f(1)=4$.
6	**(b)**	$\frac{3x}{(x-2)^2}=\frac{3}{x-2}+\frac{6}{(x-2)^2}$, integral $=3+3\ln 2$.
7	**(e)**	Quarter of a disk of radius 3: $\frac{1}{4}\pi\cdot 9=\frac{9}{4}\pi$.
8	**(d)**	$\frac{1}{3}\int_1^4(4+4x^{-2})dx=\frac{1}{3}\cdot 15=5$.
9	**(a)**	$u=\cos^{-1}x$: $\int_{\pi/4}^{\pi/2}u\,du=\frac{1}{2}\left(\frac{\pi^2}{4}-\frac{\pi^2}{16}\right)=\frac{3\pi^2}{32}$.
10	**(c)**	$\frac{1}{3}\operatorname{arcsec}\frac{2x}{3}$ from 3 to $\sqrt{3}$: $\frac{\pi}{18}-\frac{\pi}{9}=-\frac{\pi}{18}$.
11	**(a)**	$f\geq 0$ only on $[-2,2]$, so $a=-2$, $b=2$ and $a+2b=2$.
12	**(b)**	$u=1+\sqrt{x}$: $\int_4^5 20u^{-2}du=\left[-\frac{20}{u}\right]_4^5=1$.
13	**(b)**	Meet at $y=-2,1$: $\int_{-2}^1(2-y-y^2)dy=\frac{9}{2}$.
14	**(d)**	$u=1-x$: $-15\int(u^{1/2}-u^{3/2})du=6u^{5/2}-10u^{3/2}+C$.